DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-400-01 — Overhaul pump P-101A (mechanical seal and bearings)

Area	A400 Pumps	**Task**	PO Pump overhaul
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	P-101A — P-101A centrifugal pump — E 154'-0" N -18'-6" EL 100'-9"		
Equipment	P-101A	**Lines**	6"-P150-P-1001, 4"-P300-P-1003, 3/4"-P150-W-3017
Isometrics	ISO-P-1001, ISO-P-1003, ISO-W-3017 (owner's drawings, attached)		
Permits	General work permit, Lifting plan		

Reason

Mechanical seal leaking; overhaul seal and bearings.

Scope of work

1. Isolate P-101A electrically and per the isolation list; drain the casing.
2. Record as-found shaft alignment; remove the coupling guard and spacer.
3. Break the suction and discharge nozzle joints; remove the pump to the workshop.
4. Replace the mechanical seal and the bearings; inspect the impeller and wear rings.
5. Reinstall; new nozzle gaskets and studs tightened per SW-WO-5; align to the owner's alignment criteria.
6. Remove blinds and isolations; run test.

Affected items: P-101A

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1001	6"-P150-P-1001	6"-P150-P-1001	5	agrees with the model
ISO-P-1003	4"-P300-P-1003	4"-P300-P-1003	3	agrees with the model
ISO-W-3017	3/4"-P150-W-3017	3/4"-P150-W-3017	1	agrees with the model

ISO-P-1001 (PDF): ../isometrics-imported/ISO-P-1001.pdf · ISO-P-1003 (PDF): ../isometrics-imported/ISO-P-1003.pdf · ISO-W-3017 (PDF): ../isometrics-imported/ISO-W-3017.pdf — the sheets that hold the work follow the ITP.

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
P-101A	equipment	P-101A centrifugal pump		E 154'-0" N -18'-6" EL 100'-9"
P-101A-M	motor	Motor of P-101A		E 154'-0" N -18'-6" EL 100'-9"
6"-P150-P-1001/B7	item	VALVE, GATE, WEDGE, OS&Y 6" P150	6"-P150-P-1001	E 154'-0" N -13'-5-1/8" EL 102'-0"
6"-P150-P-1001/B5	item	FLANGE WELD-NECK 6" P150 RAISED FACE	6"-P150-P-1001	E 154'-0" N -14'-2-1/4" EL 102'-0"
4"-P300-P-1003/A5	item	VALVE, GATE, WEDGE, OS&Y 3" P300	4"-P300-P-1003	E 154'-0" N -16'-6" EL 105'-2-3/16"
4"-P300-P-1003/A2	item	VALVE, SWING CHECK 3" P300	4"-P300-P-1003	E 154'-0" N -16'-6" EL 104'-1/8"
3/4"-P150-W-3017/A3	item	VALVE, BALL, FLOATING 3/4" P150	3/4"-P150-W-3017	E 156'-0" N -4'-0" EL 123'-7-11/16"
3/4"-P150-W-3017/A6	item	FLANGE WELD-NECK 3/4" P150 RAISED FACE	3/4"-P150-W-3017	E 156'-0" N -4'-0" EL 124'-4-5/8"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	electrical	P-101A-M		Motor of P-101A: isolated, locked and tried at the starter			
2	valve	6"-P150-P-1001/B7	6"-P150-P-1001	Close and lock VALVE, GATE, WEDGE, OS&Y 6" P150 (6"-P150-P-1001/B7) on P-101A-S	E 154'-0" N -13'-5-1/8" EL 102'-0"		
3	blind	6"-P150-P-1001/B5	6"-P150-P-1001	Fit a blind between 6"-P150-P-1001/B7 and 6"-P150-P-1001/B5, P-101A side	E 154'-0" N -14'-4-3/16" EL 102'-0"		
4	valve	4"-P300-P-1003/A5	4"-P300-P-1003	Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (4"-P300-P-1003/A5) on P-101A-D	E 154'-0" N -16'-6" EL 105'-2-3/16"		
5	blind	4"-P300-P-1003/A2	4"-P300-P-1003	Fit a blind between 4"-P300-P-1003/A5 and 4"-P300-P-1003/A2, P-101A side	E 154'-0" N -16'-6" EL 103'-5"		

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
6	valve	3/4"-P150-W-3017/A3	3/4"-P150-W-3017	Close and lock VALVE, BALL, FLOATING 3/4" P150 (3/4"-P150-W-3017/A3) on P-101A-F	E 156'-0" N -4'-0" EL 123'-7-11/16"		
7	blind	3/4"-P150-W-3017/A6	3/4"-P150-W-3017	Fit a blind between 3/4"-P150-W-3017/A3 and 3/4"-P150-W-3017/A6, P-101A side	E 156'-0" N -4'-0" EL 124'-2-3/8"		

Parts

Description	Qty	Replaces
GASKET 4" P150 COMPRESSED SHEET	1	6"-P150-P-1001/B1
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	1	6"-P150-P-1001/B3
GASKET 3" P300 COMPRESSED SHEET	1	4"-P300-P-1003/A1
STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	1	4"-P300-P-1003/A3
Mechanical seal kit for P-101A	1	
Bearing set for P-101A	1	

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
6"-P150-P-1001/B1	4 × 1" studs (6"-P150-P-1001/B3)	35,343	3	
4"-P300-P-1003/A1	4 × 0.875" studs (4"-P300-P-1003/A3)	27,059	3	

Test — Run test

Description	Running at duty: no seal leak; bearing temperature and vibration within the owner's limits
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

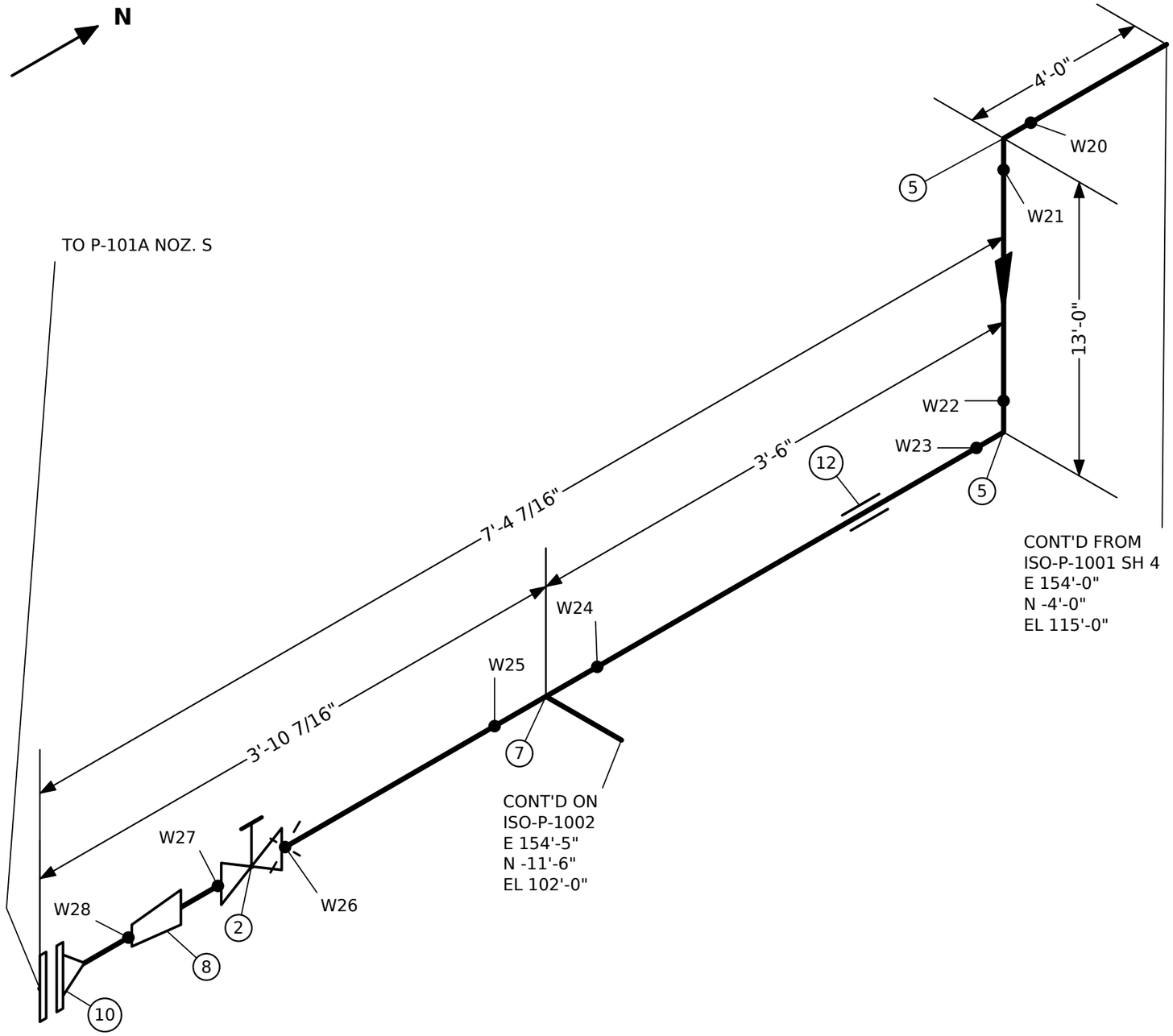
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	As-found alignment recorded	Owner alignment procedure	Recorded	Alignment record	W	R	—				
3	Seal and bearings replaced; impeller and wear rings inspected	Owner pump procedure	Clearances within the owner's criteria	Pump overhaul record	S	R	—				
4	Nozzle joints remade with new gaskets; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
5	Final alignment	Owner alignment procedure	Within the owner's alignment criteria	Alignment record	W	W	—				
6	Run test at duty	SW-TEST-2	No seal leak; temperatures and vibration within owner limits	Run test record	W	H	—				
7	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
8	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
6"-P150-P-1001/B7	isolation: valve	ISO-P-1001	5	
6"-P150-P-1001/B5	isolation: blind	ISO-P-1001	5	
4"-P300-P-1003/A5	isolation: valve	ISO-P-1003	1	
4"-P300-P-1003/A2	isolation: blind	ISO-P-1003	1	
3/4"-P150-W-3017/A3	isolation: valve	ISO-W-3017	1	
3/4"-P150-W-3017/A6	isolation: blind	ISO-W-3017	1	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	6"	GASKET 6" P150 COMPRESSED SHEET
2	1	6"	VALVE, GATE, WEDGE, OS&Y 6" P150
3	2	6"	STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P150
4	2	6"	FLANGE WELD-NECK 6" P150 RAISED FACE
5	2	6"	ELBOW 90° R = 5/3 OD 6" P150
6	17'-4 1/8"	6"	PIPE 6" P150 BEVELLED ENDS
7	1	6"	TEE EQUAL 6" P150
8	1	6"x4"	REDUCER CONCENTRIC 6" × 4" P150
9	1	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150
10	1	4"	FLANGE WELD-NECK 4" P150 RAISED FACE
11	1	4"	GASKET 4" P150 COMPRESSED SHEET
12	1	6"	TRUNNION FOR 6" LINE (TRUNNION-6x21.0)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

NOTES:

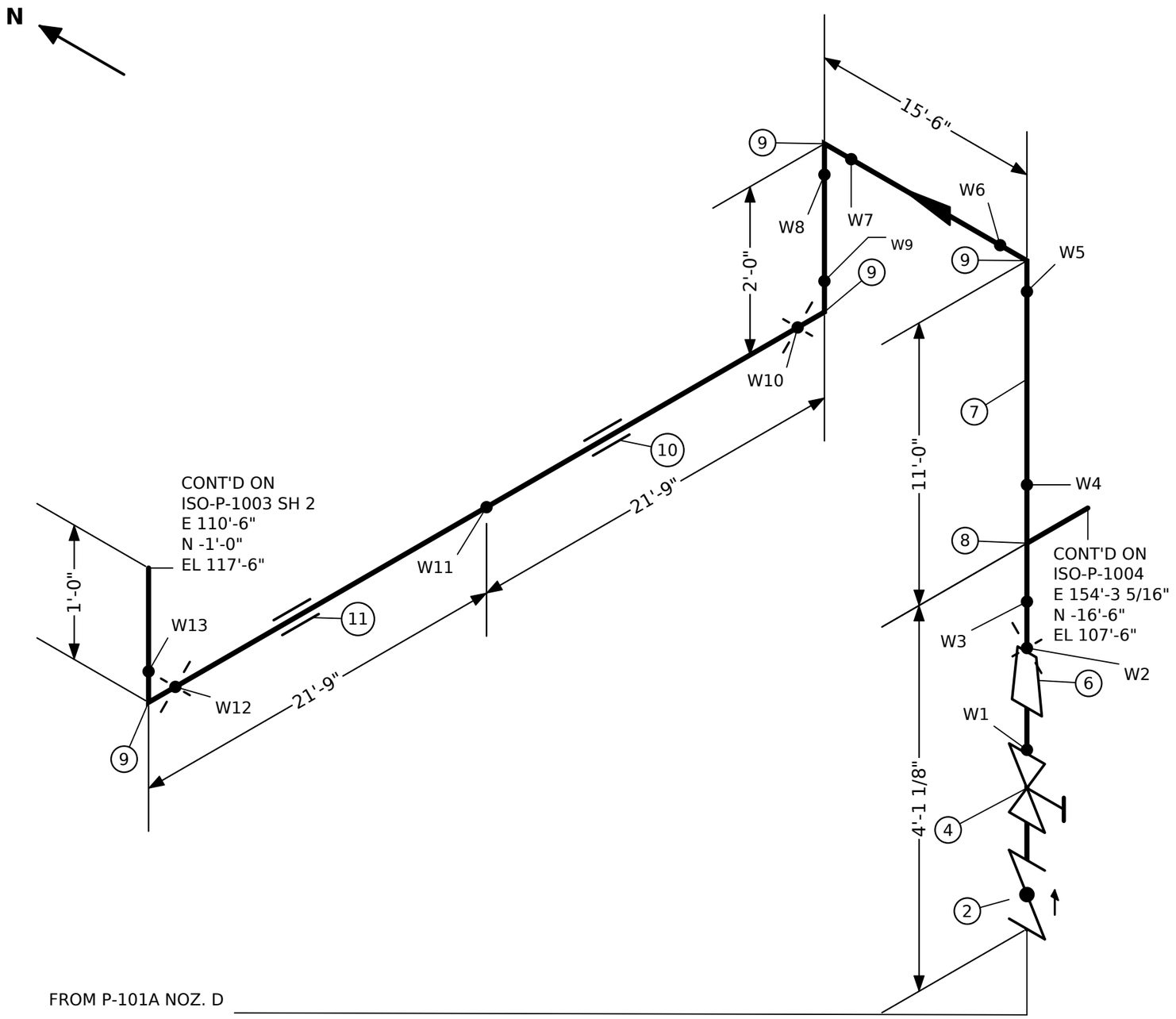
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. HYDROTEST BEFORE PAINTING
4. REMOVE BURRS FROM ALL CUT ENDS

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
6"-P150-P-1001	H150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1001	5 OF 5	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	3	3"	GASKET 3" P300 COMPRESSED SHEET
2	1	3"	VALVE, SWING CHECK 3" P300
3	3	3"	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300
4	1	3"	VALVE, GATE, WEDGE, OS&Y 3" P300
5	1	3"	FLANGE WELD-NECK 3" P300 RAISED FACE
6	1	4"x3"	REDUCER CONCENTRIC 4" × 3" P300
7	68'-11 7/16"	4"	PIPE 4" P300 BEVELLED ENDS
8	1	4"	TEE EQUAL 4" P300
9	4	4"	ELBOW 90° R = 5/3 OD 4" P300
10	1	4"	ANCHOR FOR 4" LINE (ANCHOR-4)
11	1	4"	GUIDE FOR 4" LINE (GUIDE-4)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

NOTES:

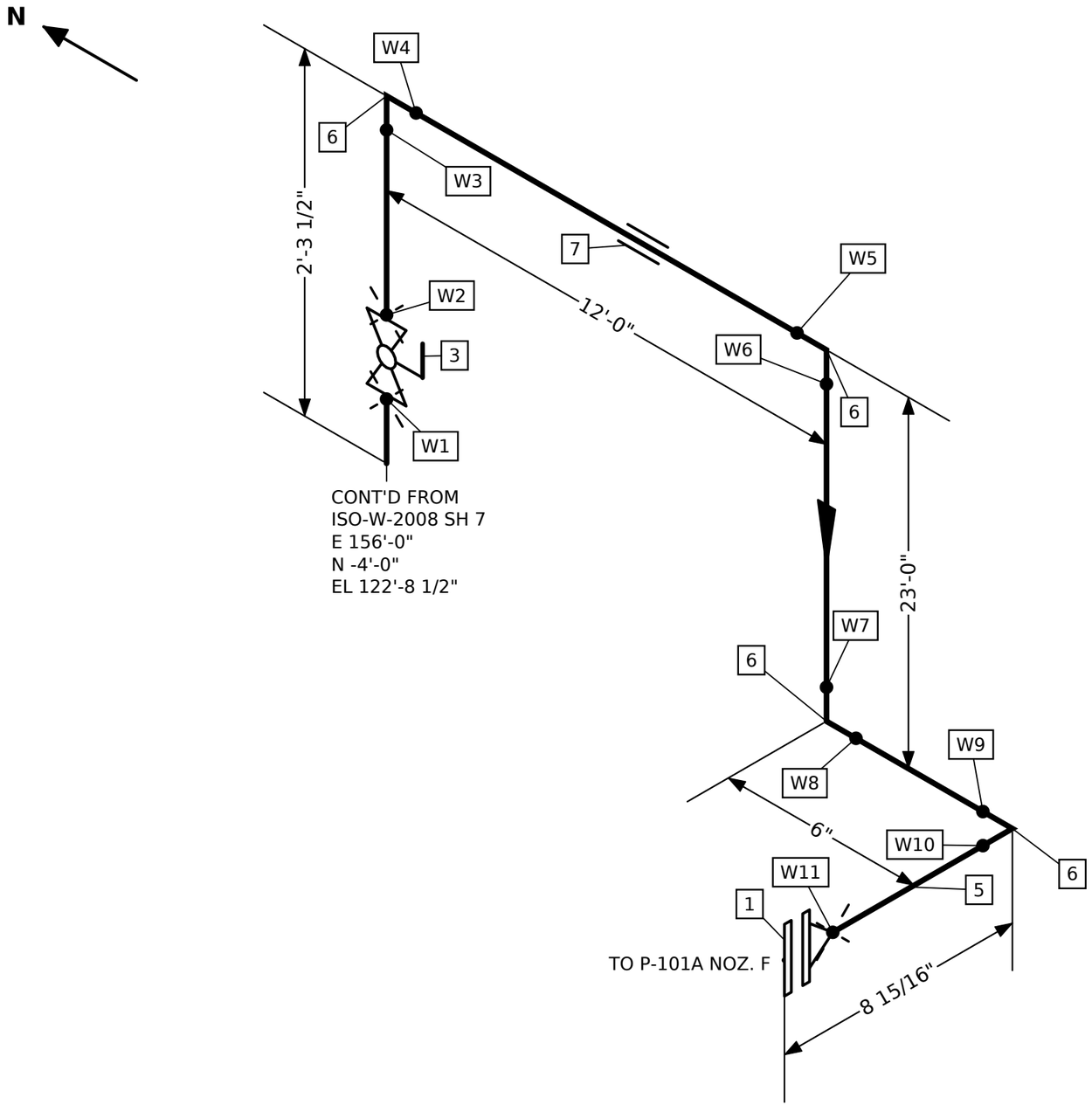
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P300-P-1003	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1003	1 OF 3	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	3	3/4"	FLANGE WELD-NECK 3/4" P150 RAISED FACE
2	3	3/4"	GASKET 3/4" P150 COMPRESSED SHEET
3	1	3/4"	VALVE, BALL, FLOATING 3/4" P150
4	3	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P150
5	35'-5 7/16"	3/4"	PIPE 3/4" P150 BEVELLED ENDS
6	4	3/4"	ELBOW 90° R = 5/3 OD 3/4" P150
7	1	3/4"	HANGER FOR 3/4" LINE (HANGER-0.75x88.0)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

COMPANY

SMARTWELD

TITLE

WATER (COOLING AND PROCESS)

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. PAINT PER PROJECT SPECIFICATION

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	W

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
3/4"-P150-W-3017	W150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-W-3017	1 OF 1	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

Notices

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

This document embeds a subset of the DejaVu Sans font (SW-LIC-3), renamed under its licence. DejaVu Sans (assets/fonts/DejaVuSans.ttf), from the DejaVu fonts 2.37 as packaged in the build environment.

Copyright (c) 2003 by Bitstream, Inc. All Rights Reserved. Bitstream Vera is a trademark of Bitstream, Inc.
DejaVu changes are in public domain.

Permission is hereby granted, free of charge, to any person obtaining a copy
of the fonts accompanying this license ("Fonts") and associated
documentation files (the "Font Software"), to reproduce and distribute the
Font Software, including without limitation the rights to use, copy, merge,
publish, distribute, and/or sell copies of the Font Software, and to permit
persons to whom the Font Software is furnished to do so, subject to the
following conditions:

The above copyright and trademark notices and this permission notice shall
be included in all copies of one or more of the Font Software typefaces.

The Font Software may be modified, altered, or added to, and in particular
the designs of glyphs or characters in the Fonts may be modified and
additional glyphs or characters may be added to the Fonts, only if the fonts
are renamed to names not containing either the words "Bitstream" or the word
"Vera".

This License becomes null and void to the extent applicable to Fonts or Font
Software that has been modified and is distributed under the "Bitstream
Vera" names.

The Font Software may be sold as part of a larger software package but no
copy of one or more of the Font Software typefaces may be sold by itself.

THE FONT SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS
OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTIES OF MERCHANTABILITY,
FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT OF COPYRIGHT, PATENT,
TRADEMARK, OR OTHER RIGHT. IN NO EVENT SHALL BITSTREAM OR THE GNOME
FOUNDATION BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, INCLUDING
ANY GENERAL, SPECIAL, INDIRECT, INCIDENTAL, OR CONSEQUENTIAL DAMAGES,
WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF
THE USE OR INABILITY TO USE THE FONT SOFTWARE OR FROM OTHER DEALINGS IN THE
FONT SOFTWARE.

Except as contained in this notice, the names of Gnome, the Gnome
Foundation, and Bitstream Inc., shall not be used in advertising or
otherwise to promote the sale, use or other dealings in this Font Software
without prior written authorization from the Gnome Foundation or Bitstream
Inc., respectively. For further information, contact: fonts at gnome dot
org.